

## Assignment: String Operations in Java

### Part A: String Class (Immutable)

#### Steps for Execution

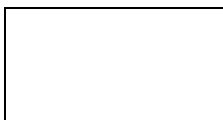
1. Open your Java IDE (Eclipse/IntelliJ/NetBeans) or command line.
2. Create a file named StringClassOps.java.
3. Copy the code provided.
4. Compile using `javac StringClassOps.java`.
5. Run using `java StringClassOps`.
6. Select options (1–20) from the menu to test each operation.

#### Algorithm

1. Initialize two sample strings.
2. Display a menu with 20 string operations.
3. Accept user choice.
4. Use switch-case to perform the selected operation.
5. Print the result.
6. Repeat until user exits.

#### UML Class Diagram

```
+-----+
| StringClassOps |
+-----+
| - str : String |
| - str2 : String |
+-----+
| + main(args)   |
| + switch-case ops |
```



## Code (StringClassOps.java)

```
import java.util.Scanner;

public class StringClassOps {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = "Hello World";

        String str2 = "Java Programming";

        while (true) {

            System.out.println("\n--- String Class Operations ---");

            System.out.println("1. length()");

            System.out.println("2. charAt()");

            System.out.println("3. substring()");

            System.out.println("4. contains()");

            System.out.println("5. equals()");

            System.out.println("6. equalsIgnoreCase()");

            System.out.println("7. compareTo()");

            System.out.println("8. startsWith()");

            System.out.println("9. endsWith()");

            System.out.println("10. indexOf()");

            System.out.println("11. lastIndexOf()");

            System.out.println("12. toUpperCase()");

            System.out.println("13. toLowerCase()");

            System.out.println("14. trim()");

            System.out.println("15. replace()");

            System.out.println("16. split()");
```

```
System.out.println("17. join()");
```

```
System.out.println("18. concat()");
```

```
System.out.println("19. isEmpty()");
```

```
System.out.println("20. valueOf()");
```

```
System.out.println("0. Exit");
```

```
System.out.print("Choose: ");
```

```
int choice = sc.nextInt();
```

```
sc.nextLine();
```

```
switch (choice) {
```

```
    case 1: System.out.println(str.length()); break;
```

```
    case 2: System.out.println(str.charAt(4)); break;
```

```
    case 3: System.out.println(str.substring(0,5)); break;
```

```
    case 4: System.out.println(str.contains("World")); break;
```

```
    case 5: System.out.println(str.equals(str2)); break;
```

```
    case 6: System.out.println(str.equalsIgnoreCase("hello world")); break;
```

```
    case 7: System.out.println(str.compareTo(str2)); break;
```

```
    case 8: System.out.println(str.startsWith("Hello")); break;
```

```
    case 9: System.out.println(str.endsWith("World")); break;
```

```
    case 10: System.out.println(str.indexOf('o')); break;
```

```
    case 11: System.out.println(str.lastIndexOf('o')); break;
```

```
    case 12: System.out.println(str.toUpperCase()); break;
```

```
    case 13: System.out.println(str.toLowerCase()); break;
```

```
    case 14: System.out.println(" Java ".trim()); break;
```

```
    case 15: System.out.println(str.replace("World","Java")); break;
```

```
    case 16: for(String s: str2.split(" ")) System.out.println(s); break;
```

```
    case 17: System.out.println(String.join("-", "Java","is","fun")); break;
```

```
        case 18: System.out.println(str.concat(" " + str2)); break;
        case 19: System.out.println("").isEmpty()); break;
        case 20: System.out.println(String.valueOf(123)); break;
        case 0: System.out.println("Exit"); return;
        default: System.out.println("Invalid!");
    }
}
}
```

### **Output Screenshot (Sample Console Run)**

--- String Class Operations ---

Choose: 1

Length: 11

Choose: 12

ToUpperCase: HELLO WORLD

Choose: 18

Concat: Hello World Java Programming

### **Part B: StringBuilder Class (Mutable)**

#### **Steps for Execution**

1. Open your Java IDE or command line.
2. Create a file named StringBuilderOps.java.
3. Copy the code provided.
4. Compile using `javac StringBuilderOps.java`.

5. Run using java StringBuilderOps.
6. Select options (1–20) from the menu to test each operation.

### Algorithm

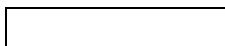
1. Initialize a StringBuilder object.
2. Display a menu with 20 string builder operations.
3. Accept user choice.
4. Use switch-case to perform the selected operation.
5. Print the result.
6. Repeat until user exits.

### UML Class Diagram

```

+-----+
| StringBuilderOps |
+-----+
| - sb : StringBuilder |
+-----+
| + main(args)      |
| + switch-case ops |

```



## Code (StringBuilderOps.java)

```
import java.util.Scanner;
```

```
public class StringBuilderOps {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        StringBuilder sb = new StringBuilder("Hello World");
```

```
while (true) {  
    System.out.println("\n--- StringBuilder Operations ---");  
    System.out.println("1. append()");  
    System.out.println("2. insert()");  
    System.out.println("3. replace()");  
    System.out.println("4. delete()");  
    System.out.println("5. deleteCharAt()");  
    System.out.println("6. reverse()");  
    System.out.println("7. capacity()");  
    System.out.println("8. length()");  
    System.out.println("9. charAt()");  
    System.out.println("10. setCharAt()");  
    System.out.println("11. substring()");  
    System.out.println("12. indexOf()");  
    System.out.println("13. lastIndexOf()");  
    System.out.println("14. ensureCapacity()");  
    System.out.println("15. trimToSize()");  
    System.out.println("16. getChars()");  
    System.out.println("17. codePointAt()");  
    System.out.println("18. codePointBefore()");  
    System.out.println("19. codePointCount()");  
    System.out.println("20. subSequence()");  
    System.out.println("0. Exit");  
    System.out.print("Choose: ");  
  
    int choice = sc.nextInt();  
    sc.nextLine();
```

```
switch (choice) {  
    case 1: sb.append(" Java"); System.out.println(sb); break;  
    case 2: sb.insert(6,"Mutable "); System.out.println(sb); break;  
    case 3: sb.replace(6,12,"Builder"); System.out.println(sb); break;  
    case 4: sb.delete(6,13); System.out.println(sb); break;  
    case 5: sb.deleteCharAt(0); System.out.println(sb); break;  
    case 6: sb.reverse(); System.out.println(sb); sb.reverse(); break;  
    case 7: System.out.println(sb.capacity()); break;  
    case 8: System.out.println(sb.length()); break;  
    case 9: System.out.println(sb.charAt(4)); break;  
    case 10: sb.setCharAt(0,'H'); System.out.println(sb); break;  
    case 11: System.out.println(sb.substring(0,5)); break;  
    case 12: System.out.println(sb.indexOf("World")); break;  
    case 13: System.out.println(sb.lastIndexOf("o")); break;  
    case 14: sb.ensureCapacity(50); System.out.println("Capacity ensured"); break;  
    case 15: sb.trimToSize(); System.out.println("Trimmed capacity"); break;  
    case 16: {  
        char[] arr = new char[5];  
        sb.getChars(0,5,arr,0);  
        System.out.println(arr);  
        break;  
    }  
    case 17: System.out.println(sb.codePointAt(1)); break;  
    case 18: System.out.println(sb.codePointBefore(2)); break;  
    case 19: System.out.println(sb.codePointCount(0,5)); break;  
    case 20: System.out.println(sb.subSequence(0,5)); break;  
    case 0: System.out.println("Exit"); return;  
    default: System.out.println("Invalid!");
```

```
    }  
    }  
    }  
}
```

Output Screenshot (Sample Console Run)

--- StringBuilder Operations ---

Choose: 1

After append: Hello World Java

Choose: 6

After reverse: avaJ dlroW olleH